

5.5 VALUE CHAIN ANALYSIS

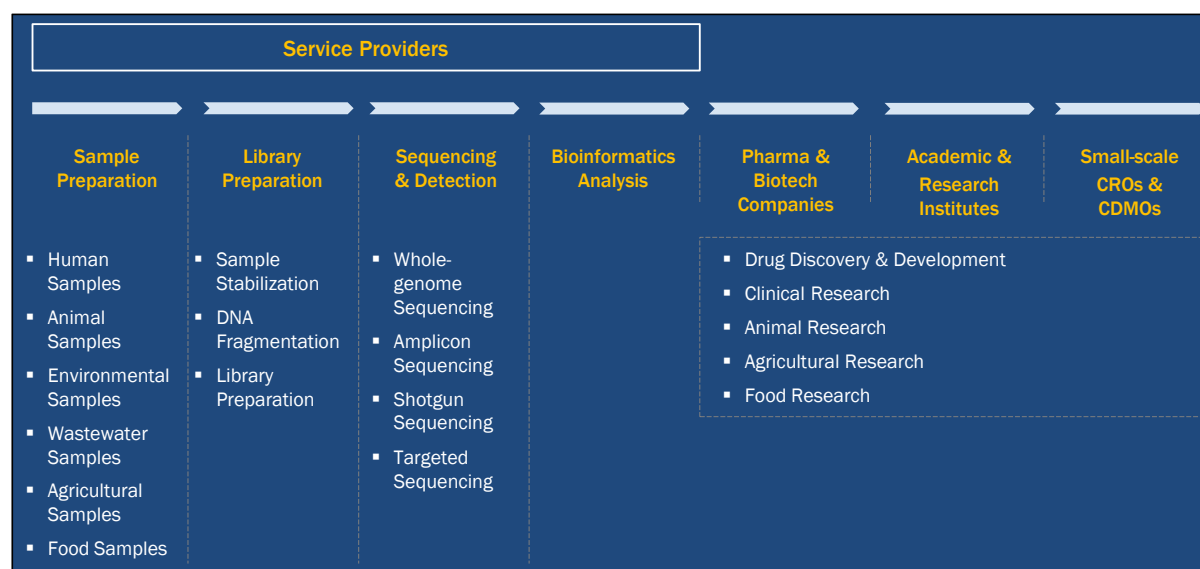
SAMPLE PREPARATION AND STUDY DESIGN SERVICES

Microbiome sequencing services start with outsourcing of study design, wherein the service provider helps in the implementation and validation of the hypothesis and the variables of interest in a sequencing project. Different variables in the aspect of study design include the number of samples required, apt sequencing technology to be used, and the requirement of follow-up studies. Service providers have expertise with respect to the collection, storage, and transport options optimized for this workflow.

SEQUENCING, LIBRARY PREPARATION, AND DATA ANALYSIS SERVICES

NGS instruments are predominantly used for the sequencing of DNA and RNA samples. Shotgun metagenomics sequencing is offered by a majority of sequencing service providers, even for deep-sequenced samples. DNA library preparation is verified for quality and yield. Genomic DNA is fragmented at specific sizes suitable for the purification/separation of desired proteins and nucleic acids. The selected fragments are then polished, A-tailed, and ligated with the full-length adapter. NGS technology is employed to sequence the sample, and the final stage involves bioinformatics analysis.

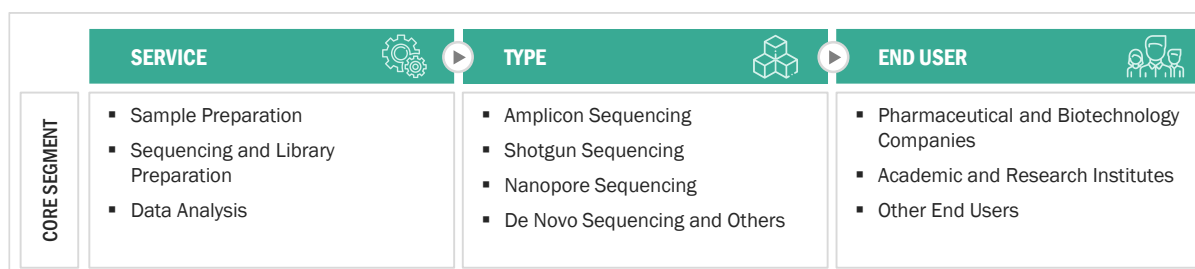
FIGURE 21 VALUE CHAIN ANALYSIS OF MICROBIOME SEQUENCING SERVICES MARKET



Source: MarketsandMarkets Analysis

5.6 ECOSYSTEM MARKET MAP

FIGURE 22 ECOSYSTEM OF MICROBIOME SEQUENCING SERVICES MARKET



Source: MarketsandMarkets Analysis

5.7 PORTER'S FIVE FORCES ANALYSIS

TABLE 6 MICROBIOME SEQUENCING SERVICES MARKET: PORTER'S FIVE FORCES ANALYSIS

PORTER'S FIVE FORCES	IMPACT
Intensity of Competitive Rivalry	Low
Bargaining Power of Suppliers	Moderate
Bargaining Power of Buyers	Moderate
Threat of Substitutes	Low to moderate
Threat of New Entrants	Low to moderate

Source: Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

5.7.1 THREAT OF NEW ENTRANTS

The global microbiome sequencing services market is not a consolidated market. The well-established players operating in this market have narrow offerings, and a very small part of their portfolios are dedicated to microbiome sequencing. However, strong distribution channels and a global presence have made it easier for these players to maintain their market position. Moreover, existing product manufacturers have strong relationships with their customers (service providers) and are engaged in long-term contracts to supply the necessary equipment. As the industry demands high capital investments for R&D, testing, and manufacturing products, the entry barriers for new service providers in the market are estimated to be low to moderate.

5.7.2 THREAT OF SUBSTITUTES

The impact of this factor is low to moderate since there are fewer substitutes for established technologies in this market. The preference for traditional technologies for sequencing microbiomes may decrease due to the availability of advanced technology-oriented services.

5.7.3 BARGAINING POWER OF SUPPLIERS

The bargaining power of suppliers is moderate in this market. This market has witnessed significant technological innovations in developing affordable and user-friendly products. Thus, suppliers have better bargaining powers if their products are highly differentiated and endowed with breakthrough technologies. Products with a high degree of technological capabilities can command a premium price in the market. The high degree of technological capabilities, coupled with a high buyer-to-seller ratio, significantly increases the bargaining power of suppliers. Furthermore, compatibility issues between instruments and consumables (which are responsible for recurrent costs) increase the bargaining power of suppliers.